Main Vocabulary on Causal Inference and ML - Week03

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Core ideas

- **Potential outcomes**:
$$Y(0), \quad Y(1),$$
the untreated and treated outcomes that each unit could have. *Example: For a patient,* $Y(1)$ *is health outcome if given drug,* $Y(0)$ *if given placebo.*

- **Causal effect**:
$$Y(1) - Y(0),$$
the change in outcome caused by switching treatment status. *Individual-level effect measuring "what would happen if this unit got treatment vs. no treatment".*

- **Average treatment effect (ATE)**:
$$\tau = \mathbb{E}[Y(1) - Y(0)].$$
The average causal effect of treatment on the population. *Population average: "On average, does treatment help everyone?"*

- **Counterfactual**: the unobserved potential outcome under the alternative treatment state. *"What would have happened if this treated patient had not received the drug?" We never observe both realities.*

- **Fundamental problem of causal inference**: for each unit, we observe only one of $Y(0)$ or $Y(1)$, never both. *No direct comparison possible within unit; must infer across units.*

- **Missing data problem**: causal inference is hard because the counterfactual outcome is always missing. *Every unit has 50% "missing data" on the other treatment outcome.*

Units and variables

- **Unit of analysis**: the entity being studied, such as a person, firm, country, or village. *Example: individual patient, company, or classroom.*
- **Outcome variable**: the response variable Y. *Example: wage, test score, survival time.*
- **Treatment variable**: the intervention variable

$$W \in \{0, 1\}.$$

  Example: $W = 1$ for drug administered, $W = 0$ for placebo.

- **Covariates**: pre-treatment variables

$$X = (X_1, \ldots, X_p).$$

  Example: age, gender, income, prior health status.

- **Observed outcome**:

$$Y = WY(1) + (1 - W)Y(0).$$

  Switch picks which potential outcome we see.

Effects and targets

- **Individual treatment effect**:

$$\tau_i = Y_i(1) - Y_i(0).$$

  Varies across units; rarely directly observed.

- **Average treatment effect (ATE)**:

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)].$$

  Overall policy question: "Should we treat everyone?"

- **Conditional average treatment effect (CATE)**:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

  "Treatment effect for people like you (with covariates x)."

- **Treatment heterogeneity**: treatment effects vary with X, so $\tau(x)$ is not constant. *Example: drug helps young patients but harms elderly.*
- **Parameter**: the target quantity being estimated, such as τ. *ATE, CATE, etc., are specific examples.*

Confounding and identification

- **Confounding**: treatment assignment depends on covariates that also affect the outcome. *Example: sicker patients get better doctors (confounder: health).*
- **Confounder**: a variable associated with both W and Y. *Example: motivation affects both job training (W) and wage (Y).*
- **Selection bias**: bias arising when treated and control groups are systematically different before treatment. *"Apples to oranges" comparison.*
- **Unconfoundedness**:
$$(Y(0), Y(1)) \perp W \mid X.$$
Treatment is as good as random after conditioning on X. *Key assumption: X captures all confounders.*
- **Selection on observables**: another name for unconfoundedness. *All relevant confounders are measured in X.*
- **Identification**:
$$\tau = \mathbb{E}[\mu_1(X) - \mu_0(X)],$$
where
$$\mu_w(x) = \mathbb{E}[Y \mid X = x, W = w].$$
ATE identifiable from observed data under unconfoundedness.

Regression adjustment

- **Regression adjustment**: estimate $\mu_0(x)$ and $\mu_1(x)$, then average their difference. *Fit separate regressions for treated/control, subtract.*
- **Conditional mean function**:
$$\mu_w(x) = \mathbb{E}[Y \mid X = x, W = w].$$
Prediction of Y within treatment group.
- **Outcome model**: a model for Y given X and W. *Linear, tree, neural net, etc.*
- **Ordinary least squares (OLS)**: a linear model used to estimate conditional means. $Y = \alpha + \beta W + \gamma X + \epsilon$.
- **Misspecification**: when the chosen model does not match the true function $\mu_w(x)$. *Linear model on nonlinear data biases results.*

Propensity scores

- **Propensity score**:
$$e(x) = \mathbb{P}(W = 1 \mid X = x).$$
This is the probability that a unit receives treatment given its covariates. In a randomized trial, this score is constant (≈ 0.5). In observational data, it can vary a lot across units. *Example: high-income more likely to get training.*

- **Propensity model**: a model used to estimate $e(x)$. This model can be simple, like logistic regression, or flexible, like a random forest. The quality of downstream causal estimation depends heavily on how well this model predicts treatment assignment.

- **Inverse propensity weighting (IPW)**:
$$\hat{\tau}_{IPW} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{W_i Y_i}{e(X_i)} - \frac{(1 - W_i) Y_i}{1 - e(X_i)} \right).$$
IPW reweights observations so that treated and control groups look more comparable. When the propensity score is correct and overlap holds, this estimator targets the ATE. But if the weights become too large, the estimator can be very unstable. *Rare types get upweighted.*

- **Importance weighting**: weighting observations by inverse probabilities. This is the general idea behind IPW. Rarely observed units receive more weight so that the sample better represents the target population.

- **Overlap**:
$$0 < e(X) < 1$$
and ideally bounded away from 0 and 1. Overlap means both treatment states are possible for units with similar covariates. Without overlap, it becomes hard or impossible to compare treated and control units fairly. *No "pure treated" or "pure control" regions.*

- **Extreme propensity score**: a value of $e(X)$ near 0 or 1, which can create unstable weights. Extreme scores are a warning sign that the data are close to violating overlap. In practice, they can inflate variance and make confidence intervals unreliable. *Trim or truncate weights to stabilize.*

Double robustness

- **Augmented inverse propensity weighting (AIPW)**:

$$\hat{\tau}_{AIPW} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{W_i\{Y_i - \hat{\mu}_1(X_i)\}}{\hat{e}(X_i)} - \frac{(1-W_i)\{Y_i - \hat{\mu}_0(X_i)\}}{1-\hat{e}(X_i)}\right].$$

  Combines outcome regression + IPW.

- **Doubly robust**: the estimator remains valid if either the outcome model or the propensity model is correct. *"Two chances to get it right".*

- **Residual**:

$$Y - \hat{\mu}_W(X).$$

  Outcome prediction error.

- **Augmentation**: the correction term added to IPW. *Residual-on-propensity reduces variance.*

- **Semiparametric efficiency**: near-optimal precision in a partially nonparametric model. *AIPW achieves efficiency bound.*

Inference and sampling

- **Central limit theorem (CLT)**: averages of many independent terms are approximately normal. *Justifies normal confidence intervals.*

- **Asymptotic normality**:

$$\sqrt{n}\,(\hat{\tau} - \tau) \xrightarrow{d} \mathcal{N}(0, V).$$

  Estimator distribution for large n.

- **Standard error**: the estimated standard deviation of an estimator. $\widehat{\mathrm{SE}} = \sqrt{\widehat{\mathrm{Var}}/n}$.

- **Confidence interval**:

$$\hat{\tau} \pm z_{1-\alpha/2} \cdot \widehat{\mathrm{SE}}(\hat{\tau}).$$

  95% CI: $\hat{\tau} \pm 1.96 \cdot \mathrm{SE}$.

- **Cross-fitting**: sample splitting so nuisance functions are trained on one fold and evaluated on another. *Avoids overfitting bias in ML.*

Machine learning and inference

- **Machine learning**: flexible prediction methods used to estimate nuisance functions such as $\mu_w(x)$ and $e(x)$. *Random forests, gradient boosting for high-dimensional X.*
- **Predictive black box**: a flexible model used mainly for prediction. *No need to interpret coefficients, just predict well.*
- **Root mean squared error (RMSE)**:

$$\text{RMSE} = \sqrt{\mathbb{E}\big[(\hat{f}(X) - f(X))^2\big]}.$$

  Primary metric for nuisance model quality.
- **Finite-sample bias**: bias present at realistic sample sizes. *ML overfitting leads to bias in causal estimates.*
- **Consistency**:

$$\hat{\theta} \xrightarrow{p} \theta$$

  as $n \to \infty$. *Large-sample unbiasedness.*

Common shortnames

Shortname	**Meaning**
ATE	Average treatment effect
CATE	Conditional average treatment effect
RCT	Randomized controlled trial
OLS	Ordinary least squares
IPW	Inverse propensity weighting
AIPW	Augmented inverse propensity weighting
PS	Propensity score
CLT	Central limit theorem
RMSE	Root mean squared error
SE	Standard error
CI	Confidence interval
ML	Machine learning